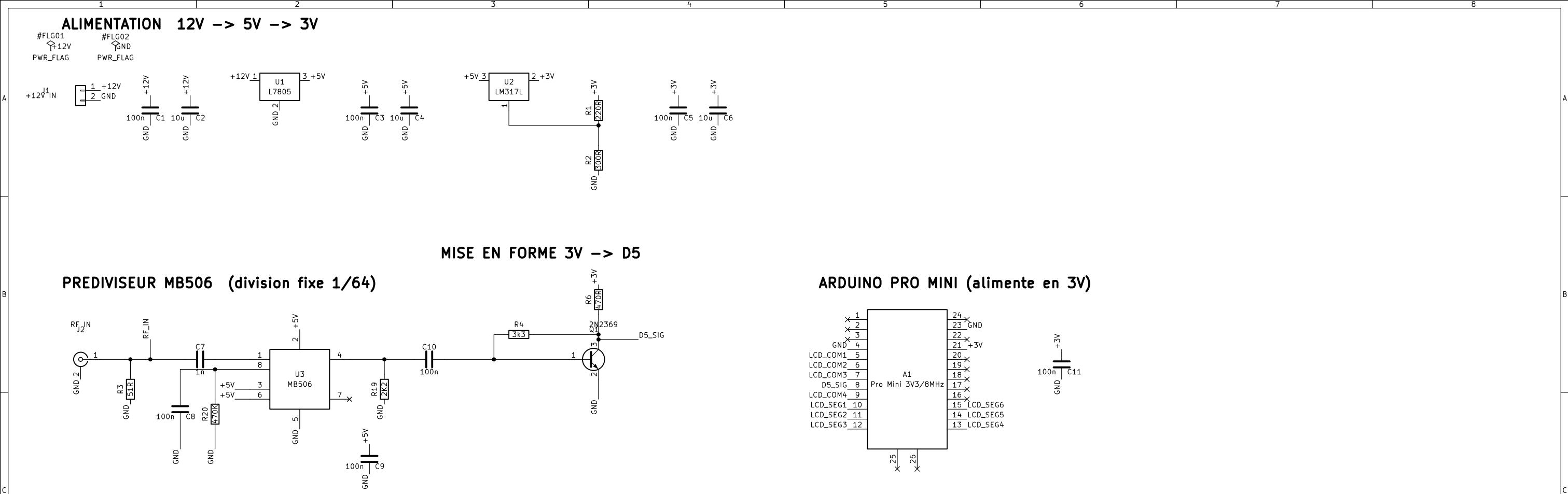


The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated down to 5V by an L7805 (U1) and then to 3V by an LM317L (U2). Various capacitors (C1, C2, C3, C4, C5, C6) are used for decoupling and filtering at different stages. Resistors R1 (220Ω) and R2 (600Ω) are also shown in the 3V line.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning for the D5 pin. A 3V signal is connected to the D5 pin of the Arduino Pro Mini (A1) through a 220Ω resistor (R1). The signal is also connected to a 3V pin of the Arduino Pro Mini (A1) through a 600Ω resistor (R2).
- PREDIVISEUR MB506 (division fixe 1/64):** This section shows the signal conditioning for the MB506 sensor. The sensor's output is connected to the D5 pin of the Arduino Pro Mini (A1) through a 220Ω resistor (R1). The sensor is powered by a 5V supply and has a 100nF capacitor (C1) connected to its ground.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini (A1) connected to the 3V supply. The Arduino's GND is connected to the common ground of the circuit. The Arduino's VCC is connected to the 3V supply. The Arduino's D5 pin is connected to the D5 signal line. The Arduino's D6 pin is connected to the D6 signal line.



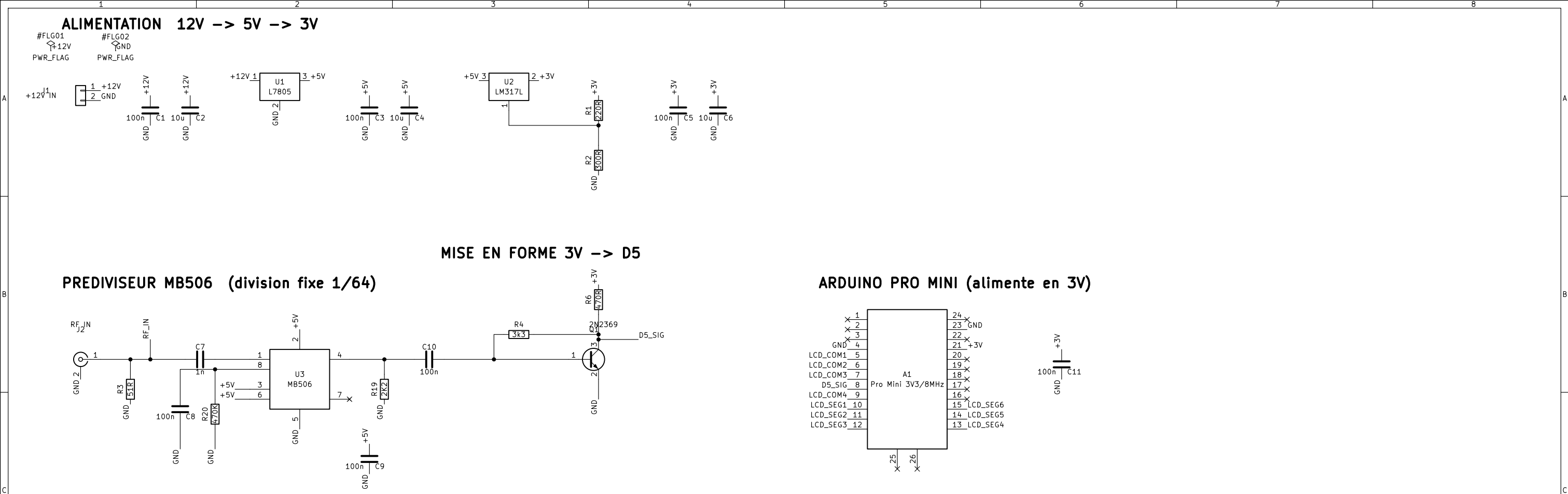
The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated down to 5V by an L7805 (U1) and then to 3V by an LM317L (U2). Various capacitors (C1, C2, C3, C4, C5, C6) are used for decoupling and filtering at different stages. Resistors R1 (220Ω) and R2 (600Ω) are also present in the 3V line.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning for the D5 pin. A 3V signal is connected to the D5 pin of the Arduino Pro Mini (A1) through a 220Ω resistor (R1). The signal is also connected to the D5 pin of the MB506 (U3) through a 220Ω resistor (R2). The MB506 is configured as a voltage divider (1/64) and its output is connected to the D5 pin of the Arduino Pro Mini (A1) through a 220Ω resistor (R3).
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini (A1) connected to the 3V supply. The Arduino is configured to operate at 3V. The power pins (GND, +3V) are connected to the 3V supply. The signal pins (D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24) are connected to the corresponding pins of the MB506 (U3).

The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated down to 5V by an L7805 (U1) and then to 3V by an LM317L (U2). Various capacitors (C1, C2, C3, C4, C5, C6) are used for decoupling and filtering at different stages. Resistors R1 (220Ω) and R2 (500Ω) are also present in the 3V line.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning for the D5 pin. A 3V signal is connected to the D5 pin of the Arduino Pro Mini (A1) through a 220Ω resistor (R1). The signal is also connected to a 5V pin through a 500Ω resistor (R2).
- PREDIVISEUR MB506 (division fixe 1/64):** This section shows a fixed divider circuit. A 5V signal is connected to the input of the MB506 (U3) through a 100nF capacitor (C1). The output of the divider is connected to the D5 pin of the Arduino Pro Mini (A1) through a 100nF capacitor (C2). The divider is also connected to a 5V pin through a 100nF capacitor (C3).

The Arduino Pro Mini (A1) is shown with its pin connections for power (GND, +5V, +3V) and signal (D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24).

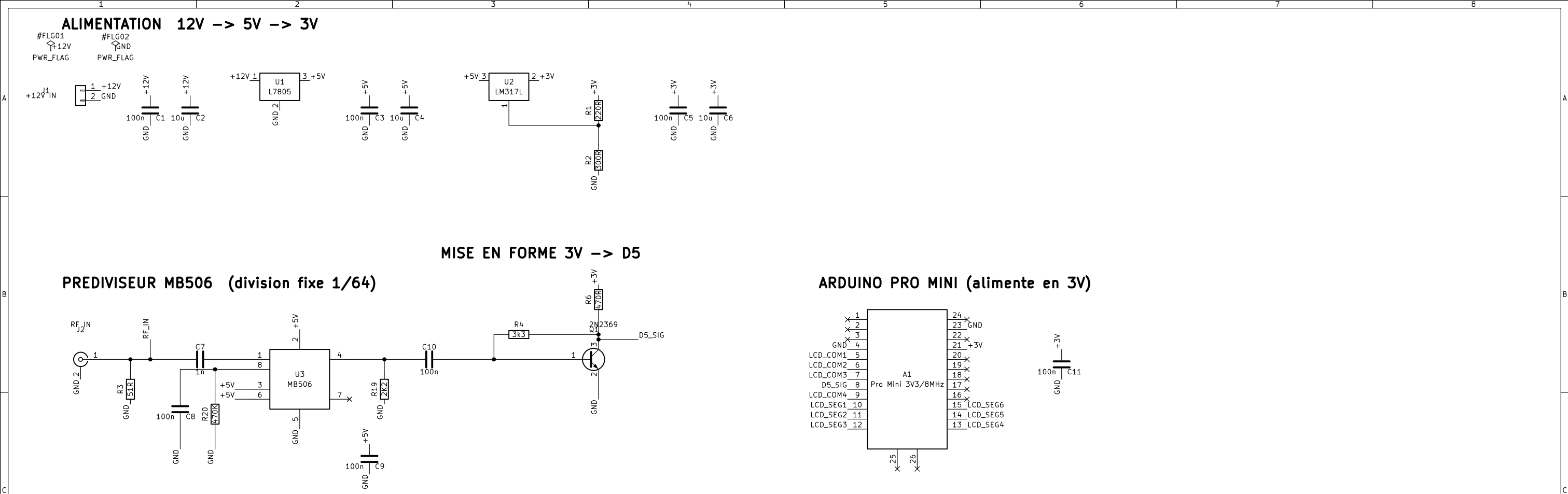


The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input (PWR_FLAG) connected to the input of a voltage divider (U1, L7805) and a 100nF capacitor (C1). The output of the divider is 5V, which is connected to the input of a second voltage divider (U2, LM317L) and a 100nF capacitor (C3). The output of the second divider is 3V, which is connected to the input of a third voltage divider (U3, MB506) and a 100nF capacitor (C5). The output of the third divider is 1.2V, which is connected to the input of a fourth voltage divider (U4, MB506) and a 100nF capacitor (C7). The output of the fourth divider is 0.3V, which is connected to the input of a fifth voltage divider (U5, MB506) and a 100nF capacitor (C9). The output of the fifth divider is 0.075V, which is connected to the input of a sixth voltage divider (U6, MB506) and a 100nF capacitor (C11). The output of the sixth divider is 0.01875V, which is connected to the input of a seventh voltage divider (U7, MB506) and a 100nF capacitor (C13). The output of the seventh divider is 0.0046875V, which is connected to the input of an eighth voltage divider (U8, MB506) and a 100nF capacitor (C15). The output of the eighth divider is 0.001171875V, which is connected to the input of a ninth voltage divider (U9, MB506) and a 100nF capacitor (C17). The output of the ninth divider is 0.00029296875V, which is connected to the input of a tenth voltage divider (U10, MB506) and a 100nF capacitor (C19). The output of the tenth divider is 7.32421875e-05V, which is connected to the input of an eleventh voltage divider (U11, MB506) and a 100nF capacitor (C21). The output of the eleventh divider is 1.8310546875e-05V, which is connected to the input of a twelfth voltage divider (U12, MB506) and a 100nF capacitor (C23). The output of the twelfth divider is 4.57763671875e-06V, which is connected to the input of a thirteenth voltage divider (U13, MB506) and a 100nF capacitor (C25). The output of the thirteenth divider is 1.1444091796875e-06V, which is connected to the input of a fourteenth voltage divider (U14, MB506) and a 100nF capacitor (C27). The output of the fourteenth divider is 2.86102294921875e-07V, which is connected to the input of a fifteenth voltage divider (U15, MB506) and a 100nF capacitor (C29). The output of the fifteenth divider is 7.152557373046875e-08V, which is connected to the input of a sixteenth voltage divider (U16, MB506) and a 100nF capacitor (C31). The output of the sixteenth divider is 1.78813934326171875e-08V, which is connected to the input of a seventeenth voltage divider (U17, MB506) and a 100nF capacitor (C33). The output of the seventeenth divider is 4.470348358154296875e-09V, which is connected to the input of an eighteenth voltage divider (U18, MB506) and a 100nF capacitor (C35). The output of the eighteenth divider is 1.11758708953857421875e-09V, which is connected to the input of a nineteenth voltage divider (U19, MB506) and a 100nF capacitor (C37). The output of the nineteenth divider is 2.79396772384643546875e-10V, which is connected to the input of a twentieth voltage divider (U20, MB506) and a 100nF capacitor (C39). The output of the twentieth divider is 6.984919309616088671875e-11V, which is connected to the input of a twenty-first voltage divider (U21, MB506) and a 100nF capacitor (C41). The output of the twenty-first divider is 1.74622982740402216796875e-11V, which is connected to the input of a twenty-second voltage divider (U22, MB506) and a 100nF capacitor (C43). The output of the twenty-second divider is 4.36557456851005541746875e-12V, which is connected to the input of a twenty-third voltage divider (U23, MB506) and a 100nF capacitor (C45). The output of the twenty-third divider is 1.091393642127513854375e-12V, which is connected to the input of a twenty-fourth voltage divider (U24, MB506) and a 100nF capacitor (C47). The output of the twenty-fourth divider is 2.7284841053187846359375e-13V, which is connected to the input of a twenty-fifth voltage divider (U25, MB506) and a 100nF capacitor (C49). The output of the twenty-fifth divider is 6.82121026329696158984375e-14V, which is connected to the input of a twenty-sixth voltage divider (U26, MB506) and a 100nF capacitor (C51). The output of the twenty-sixth divider is 1.7053025658242403974609375e-14V, which is connected to the input of a twenty-seventh voltage divider (U27, MB506) and a 100nF capacitor (C53). The output of the twenty-seventh divider is 4.26325641456060099365625e-15V, which is connected to the input of a twenty-eighth voltage divider (U28, MB506) and a 100nF capacitor (C55). The output of the twenty-eighth divider is 1.0658141036401502484140625e-15V, which is connected to the input of a twenty-ninth voltage divider (U29, MB506) and a 100nF capacitor (C57). The output of the twenty-ninth divider is 2.66453525910037562103515625e-16V, which is connected to the input of a thirtieth voltage divider (U30, MB506) and a 100nF capacitor (C59). The output of the thirtieth divider is 6.661338147750939052587890625e-17V, which is connected to the input of a thirty-first voltage divider (U31, MB506) and a 100nF capacitor (C61). The output of the thirty-first divider is 1.66533453693773476314696875e-17V, which is connected to the input of a thirty-second voltage divider (U32, MB506) and a 100nF capacitor (C63). The output of the thirty-second divider is 4.16333634234433690786740625e-18V, which is connected to the input of a thirty-third voltage divider (U33, MB506) and a 100nF capacitor (C65). The output of the thirty-third divider is 1.0408340855860842269668515625e-18V, which is connected to the input of a thirty-fourth voltage divider (U34, MB506) and a 100nF capacitor (C67). The output of the thirty-fourth divider is 2.60208521396521056741712890625e-19V, which is connected to the input of a thirty-fifth voltage divider (U35, MB506) and a 100nF capacitor (C69). The output of the thirty-fifth divider is 6.505213034913026418542821875e-20V, which is connected to the input of a thirty-sixth voltage divider (U36, MB506) and a 100nF capacitor (C71). The output of the thirty-sixth divider is 1.62630325872825660463570546875e-20V, which is connected to the input of a thirty-seventh voltage divider (U37, MB506) and a 100nF capacitor (C73). The output of the thirty-seventh divider is 4.065758146820641511589263671875e-21V, which is connected to the input of a thirty-eighth voltage divider (U38, MB506) and a 100nF capacitor (C75). The output of the thirty-eighth divider is 1.01643953670516037789731591796875e-21V, which is connected to the input of a thirty-ninth voltage divider (U39, MB506) and a 100nF capacitor (C77). The output of the thirty-ninth divider is 2.541098841762900944743289794921875e-22V, which is connected to the input of a fortieth voltage divider (U40, MB506) and a 100nF capacitor (C79). The output of the fortieth divider is 6.3527471044072523618582244873046875e-23V, which is connected to the input of a forty-first voltage divider (U41, MB506) and a 100nF capacitor (C81). The output of the forty-first divider is 1.588186776101813090464556121826171875e-23V, which is connected to the input of a forty-second voltage divider (U42, MB506) and a 100nF capacitor (C83). The output of the forty-second divider is 3.9704669152545327261613903045654296875e-24V, which is connected to the input of a forty-third voltage divider (U43, MB506) and a 100nF capacitor (C85). The output of the forty-third divider is 9.9261672881363318154034757614140625e-25V, which is connected to the input of a forty-fourth voltage divider (U44, MB506) and a 100nF capacitor (C87). The output of the forty-fourth divider is 2.481541822034082953850868940353515625e-25V, which is connected to the input of a forty-fifth voltage divider (U45, MB506) and a 100nF capacitor (C89). The output of the forty-fifth divider is 6.2038545550852073846271723508837890625e-26V, which is connected to the input of a forty-sixth voltage divider (U46, MB506) and a 100nF capacitor (C91). The output of the forty-sixth divider is 1.550963638771301846156793087720946875e-26V, which is connected to the input of a forty-seventh voltage divider (U47, MB506) and a 100nF capacitor (C93). The output of the forty-seventh divider is 3.8774090969282546153919827193023671875e-27V, which is connected to the input of a forty-eighth voltage divider (U48, MB506) and a 100nF capacitor (C95). The output of the forty-eighth divider is 9.69352274232063653847745679825591796875e-28V, which is connected to the input of a forty-ninth voltage divider (U49, MB506) and a 100nF capacitor (C97). The output of the forty-ninth divider is 2.42338068558015913461936419956397946875e-28V, which is connected to the input of a fiftieth voltage divider (U50, MB506) and a 100nF capacitor (C99). The output of the fiftieth divider is 6.05845171395039783654841049890994875e-29V, which is connected to the input of a fifty-first voltage divider (U51, MB506) and a 100nF capacitor (C101). The output of the fifty-first divider is 1.5146129284875994591371026247274871875e-29V, which is connected to the input of a fifty-second voltage divider (U52, MB506) and a 100nF capacitor (C103). The output of the fifty-second divider is 3.78653232121899864784275656181871796875e-30V, which is connected to the input of a fifty-third voltage divider (U53, MB506) and a 100nF capacitor (C105). The output of the fifty-third divider is 9.466330803047496619606891404546794921875e-31V,

The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

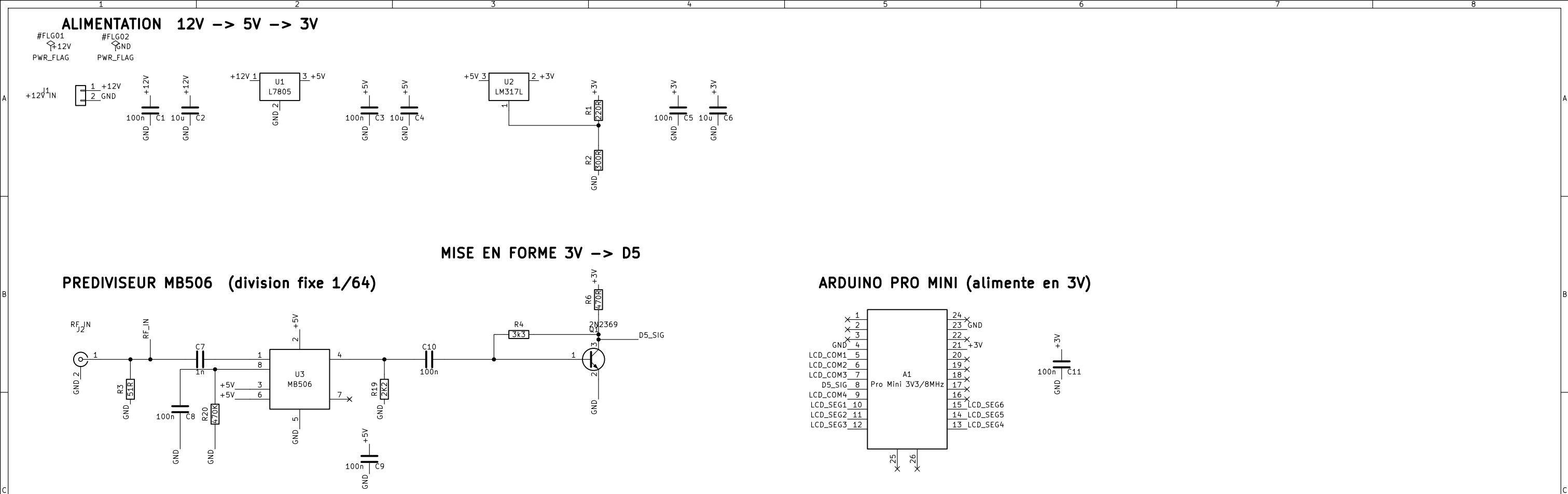
- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated to 5V by an L7805 (U1) and then to 3V by an LM317L (U2). Various capacitors (C1, C2, C3, C4, C5, C6) and resistors (R1, R2) are used for filtering and regulation.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning. A 3V input is connected to a D5_SIG pin. The signal is conditioned by a network of resistors (R3, R4, R5, R6) and capacitors (C7, C8, C9, C10) to ensure proper signal levels and timing.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini microcontroller. It is powered by the 3V supply. The pin connections for the LCD display (LCD_COM1, LCD_COM2, LCD_COM3, LCD_COM4, LCD_SEG1, LCD_SEG2, LCD_SEG3, LCD_SEG4, LCD_SEG5, LCD_SEG6) and the D5_SIG pin are shown.



The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated down to 5V by an L7805 (U1) and then to 3V by an LM317L (U2). Various capacitors (C1, C2, C3, C4, C5, C6) are used for decoupling and filtering at different stages. Resistors R1 (220Ω) and R2 (600Ω) are also present in the 3V line.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning for the D5 pin. A 3V signal is connected to the D5 pin of the Arduino Pro Mini (A1) through a 220Ω resistor (R1). The signal is also connected to a 3V pin of the Arduino Pro Mini (A1) through a 600Ω resistor (R2).
- PREDIVISEUR MB506 (division fixe 1/64):** This section shows the signal conditioning for the MB506 sensor. The sensor's output is connected to the D5 pin of the Arduino Pro Mini (A1) through a 220Ω resistor (R1). The sensor is powered by a 5V supply and has a 100nF capacitor (C1) connected to its ground.

The Arduino Pro Mini (A1) is shown with its pin connections for power (GND, +5V, +3V) and signal (D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24).



The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated by a 7805 (U1) to 5V. The 5V is then regulated by an LM317L (U2) to 3V. The 3V output is connected to the Arduino Pro Mini's 3V pin. Various capacitors (C1, C2, C3, C4, C5, C6) are used for decoupling and filtering at different stages.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning for the D5 pin. The 3V signal is connected to the D5 pin of the Arduino Pro Mini. The signal is then connected to the D5 pin of the MB506 module. The MB506 module's output is connected to the D5 pin of the Arduino Pro Mini. The module is powered by 5V and GND.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini module. It is powered by 3V and GND. The module's pins are labeled: GND, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26. The module is connected to the power supply and signal conditioning circuit.

The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated down to 5V by an L7805 (U1) and then to 3V by an LM317L (U2). Various capacitors (C1, C2, C3, C4, C5, C6) are used for decoupling and filtering at different stages. Resistors R1 (220Ω) and R2 (500Ω) are used for current limiting and voltage division.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning circuit. A 3V input is connected to a D5_SIG pin. The signal is conditioned by a network of resistors (R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100) and capacitors (C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100) to ensure proper signal levels and timing.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini board connected to the 3V supply. The board is powered by a 3V input connected to a D5_SIG pin. The board is also connected to a 12V input connected to a PWR_FLAG pin. The board is powered by a 3V input connected to a D5_SIG pin.

The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input connected to a PWR_FLAG pin. The 12V is regulated down to 5V by an L7805 (U1) and then to 3V by an LM317L (U2). Various capacitors (C1, C2, C3, C4, C5, C6) and resistors (R1, R2) are used for filtering and regulation.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning circuit. A 3V input is connected to a D5_SIG pin. The signal is conditioned by a network of resistors (R3, R4, R5, R6) and capacitors (C7, C8, C9, C10, C11) to ensure proper signal levels and timing.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini microcontroller. It is powered by the 3V supply and has various pins connected to the signal conditioning circuit. The microcontroller is labeled "Pro Mini 3V3/8MHz".

The schematic diagram illustrates the power supply and signal conditioning circuit for the Arduino Pro Mini. It is divided into three main sections:

- ALIMENTATION 12V -> 5V -> 3V:** This section shows the power supply chain. It starts with a 12V input (PWR_FLAG) connected to the input of a voltage divider (U1, L7805). The output of U1 is 5V, which is then connected to the input of a second voltage divider (U2, LM317L). The output of U2 is 3V, which is connected to the Arduino Pro Mini's 3V pin. The circuit includes several capacitors (C1, C2, C3, C4, C5, C6) for filtering and decoupling, and resistors (R1, R2) for the voltage dividers.
- MISE EN FORME 3V -> D5:** This section shows the signal conditioning circuit. It starts with a 3V input (D5_SIG) connected to the input of a voltage divider (U3, MB506). The output of U3 is 5V, which is then connected to the Arduino Pro Mini's 5V pin. The circuit includes several capacitors (C7, C8, C9, C10) for filtering and decoupling, and resistors (R3, R4, R5, R6) for the voltage dividers.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini board, which is powered by the 3V supply. The board's pins are labeled, including GND, 5V, 3V, and various digital and analog pins.